

Zhuoli Ouyang | Curriculum Vitae

Beijing – China

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Personal Information

Date of Birth: 2004-10-15

Nationality: China

GPA: 3.94/4.00 (last year), 3.88/4.00 (total)

Languages: English, Chinese, Japanese

Research Interest

I am an undergraduate student with a strong foundation in mathematics and AI. My research interests span **generative models**, **reinforcement learning**, and **optimization**, with a focus on building more **efficient AI** systems. I am particularly excited about **embodied AI** and the challenge of enabling agents to perceive, reason, and act in complex environments. I aim to bridge principled mathematical thinking with practical system design, and look forward to contributing to research at the frontier of these areas.

Education

Southern University of Science and Technology—Second-year undergraduate

Information Engineering, Dept. of Electronic & Electrical Engineering, China

GPA: 3.88, Rank: **4/46**, Honor Class

Signals and Systems (97/100) [Signal & Systems](#), Engineering Mathematics (95/100), Probability Theory and Mathematical Statistics (95/100)

Shenzhen

2024–Present

Southern University of Science and Technology—First-year undergraduate

Department of Mathematics, China

GPA: 3.86, Rank: **94/1346** (overall in university), **Top 7%**, Fields Elite Class (Honor Class)

Algebraic Geometry (Graduate level), Advanced Functional Analysis (Graduate level), Analysis (97/100, Honor) [Analysis](#), Advanced Algebra (97/100, Honor), C programming (95/100)

Shenzhen

2023–2024

Tsinghua University High School

High School Diploma, China

Beijing

2020–2023

School Affiliated to Tsinghua University

Secondary Education, China

Beijing

2017–2020

Honors & Awards

2026: ALT 2026 Best Student Paper Award

2024: SUSTech, Department of Electronics and Electrical Engineering-(Honor Class)

2023: SUSTech, Mathematics Department - Fields Elite Class-(Honor Class)

Publications

RMNP: Row-Momentum Normalized Preconditioning for Scalable Matrix-Based Optimization *arXiv 2026*

Shenyang Deng*, **Zhuoli Ouyang*** et al. (*Equal contribution)

[\[arXiv\]](#) [\[Post by Tony\]](#) [\[Post by Yushun\]](#)

Depth, Not Data: An Analysis of Hessian Spectral Bifurcation

Shenyang Deng*, **Zhuoli Ouyang***, Boyao Liao* et al. (*Equal contribution)

ISIT 2026

Strong Accept [\[arXiv\]](#)

Suspicious Alignment of SGD: A Fine-Grained Step Size Condition Analysis

Shenyang Deng*, Boyao Liao*, **Zhuoli Ouyang** et al. (*Equal contribution)

ALT 2026

Best Student Paper Award [\[arXiv\]](#)

Doloris: Dual Conditional Diffusion Implicit Bridges with Sparsity Masking Strategy for Unpaired Single-Cell Perturbation Estimation

Changci Chi, Jun Xia, Yufei Huang, **Zhuoli Ouyang** et al.

ICLR 2026

[\[OpenReview\]](#)

Design Intent Driven Multi-Agent Reinforcement Learning With Current-Transient Automatic Clustering for Analog LDO Sizing

Han Wu, Haoning Jiang, **Zhuoli Ouyang** et al.

TCAD

ACEMARL: Adaptive Clustering Enhanced Multi-Agent Reinforcement Learning for Analog Circuit Sizing

Han Wu*, Haoning Jiang*, **Zhuoli Ouyang** et al. (*Equal contribution)

DATE 2026

Research & Study Experience

Dartmouth College

Research Intern – Optimizers & Learning Theory

Advisor: Yaoqing Yang

Hanover, NH

2025–Present

Working on optimization algorithms and learning theory, contributing to theoretical analysis and algorithm design for machine learning.

Achievement: ICML 2026 (Under Review); ALT 2026 (Best Student Paper Award); ISIT 2026

AI for Scientific Simulation and Discovery Lab, Westlake University

Research Intern

Advisor: Tailing Wu

Hangzhou

2025–Present

Working on generative AI methods (diffusion models, flow matching, stochastic optimization) for scientific simulation and discovery. Research spans AI for physical sciences—including fluid dynamics, plasma control, and embodied intelligence—as well as AI for life science, focusing on generative modeling of biological systems and evolutionary mechanisms.

Shanghai AI Laboratory / OpenDataLab

Research Intern – Optimizer Benchmark & Survey

Advisors: Cheng Tan, Siyuan Li

Shanghai

2025–Present

Conducting a comprehensive benchmark and survey of optimization algorithms for large-scale AI, covering classical and modern optimizers across convergence, efficiency, and generalization.

HKUST (Guangzhou)

Research Intern – Theory-First Learning & Optimization for AI

Advisor: Yao Shu

Guangzhou

2025–Present

Working on foundational optimization and learning theory for large-scale AI systems. Research spans black-box optimization (zeroth-order, Bayesian, bandit, RL), learning dynamics (generalization, stability, robustness), and their application to LLMs and intelligent agents.

Westlake CAIRI AI Lab, Westlake University

Research Intern – Generative Modeling for Biology (AI4Science)

Advisor: Changxi Chi

Hangzhou

2025–Present

Working on generative modeling for biological systems, including diffusion models, flow matching, and bridge models for single-cell perturbation estimation and cell imaging.

Achievement: ICLR 2026

SUSTech

AI for EDA

Advisor: Junmin Jiang

Shenzhen

2025

Applying Reinforcement Learning and AI algorithms to analog circuit design.

Achievement: AAAI 2026, DATE 2026, TCAD

Dept. of Math, Fields Elite - Honor Class & Graduate Courses

Mathematics & Theoretical Physics (incl. FAU online)

2023–2025

Courses: Differential Manifolds, Algebraic Topology, Representation Theory, Abstract Algebra, Algebraic Geometry, Functional Analysis, Quantum Mechanics, General Relativity, Electromagnetism, etc.

See also: Algebraic Topology, Differential Forms in Algebraic Topology, Geometric Anatomy of Theoretical Physics

Other Skills

Programming: Python, C, C++, MATLAB, LaTeX, etc.

Software: PyTorch, PETSc, MATLAB, Fluent

Soccer: Second-class National Athlete in Football

Piano: Level 8, Central Conservatory of Music, Beijing

Hobbies: Exploring mathematics, Coding, Philosophy and History, Playing piano, Playing soccer